

```
MariaDB [(none)]> create database PROJECT;
MariaDB [(none)]> use PROJECT;
Database changed
```

```
MariaDB [PROJECT]> create table Department
-> (depthNo int primary key, depthName varchar(50), mgrEmpNo int);
```

```
MariaDB [PROJECT]> create table Employee
-> (empNo int primary key, fName varchar(50), lName varchar(50), address varchar(100),
DOB date, sex varchar(10), position varchar(50), depthNo int, foreign key (depthNo)
references Department(depthNo));
```

```
MariaDB [PROJECT]> create table Project
-> (projNo int primary key, projName varchar(50), depthNo int, foreign key (depthNo)
references Department(depthNo));
```

```
MariaDB [PROJECT]> create table WorksOn
-> (empNo int, projNo int, dateWorked date, hoursWorked decimal (5,2), primary key
(empNo, projNo, dateWorked), foreign key (empNo) references Employee (empNo), foreign
key (projNo) references Project(projNo));
```

```
MariaDB [PROJECT]> alter table Employee
```

```
-> modify sex char(1);
```

```
MariaDB [PROJECT]> alter table Employee
```

```
-> add constraint chk_sex
```

```
-> check (sex in ('M', 'F'));
```

```
MariaDB [PROJECT]> alter table Employee
```

```
-> add constraint chk_position
```

```
-> check (position in ('Manager', 'Team Leader', 'Analyst', 'Software Developer'));
```

```
MariaDB [PROJECT]> alter table WorksOn
```

```
-> add constraint chk_hoursWorked
```

```
-> check (hoursWorked between 0 and 40);
```

```
MariaDB [PROJECT]> create view femaleManagedProjects as
```

```
-> select p.projNo, p.projName, p.depthNo
```

```
-> from Project p
```

```
-> join Department d on p.depthNo = d.depthNo
```

```
-> join Employee e on d.mgrEmpNo = e.empNo
```

```
-> where e.sex= 'F'
```

```
-> order by p.projNo;
```

```
MariaDB [PROJECT]> create view EmpProjectHours as
```

```
-> select e.empNo, e.fName, e.lName, p.projName, w.hoursWorked
```

```
-> from Employee e
```

```
-> join WorksOn w on e.empNo= w.empNo
```

-> join Project p on w.projNo= p.projNo;

MariaDB [project]> INSERT INTO Department (depthNo, depthName, mgrEmpNo)

-> VALUES

-> (1, 'IT', 101),

-> (2, 'Finance', 102),

-> (3, 'Marketing', 103),

-> (4, 'HR', 104),

-> (5, 'Operations', NULL);

Query OK, 5 rows affected (0.050 sec)

Records: 5 Duplicates: 0 Warnings: 0

MariaDB [project]> INSERT INTO Employee

-> (empNo, fName, lName, address, DOB, sex, position, depthNo)

-> VALUES

-> (101, 'Ama', 'Mensah', 'Accra', '1990-05-10', 'F', 'Manager', 1),

->

-> (102, 'Kwame', 'Asare', 'Tema', '1989-03-15', 'M', 'Manager', 2),

->

-> (103, 'Akosua', 'Boateng', 'Kumasi', '1992-07-20', 'F', 'Manager', 3),

->

-> (104, 'Kofi', 'Owusu', 'Cape Coast', '1991-09-08', 'M', 'Manager', 4),

->

-> (105, 'Abena', 'Nyarko', 'Takoradi', '1993-12-01', 'F', 'Manager', 5);

Query OK, 5 rows affected (0.033 sec)

Records: 5 Duplicates: 0 Warnings: 0

MariaDB [project]> INSERT INTO Project

-> (projNo, projName, depthNo)

-> VALUES

-> (1, 'Payroll System', 2),

->

-> (2, 'Inventory App', 1),

->

-> (3, 'Online Banking', 2),

->

-> (4, 'Student Portal', 3),

->

-> (5, 'HR Management', 4);

Query OK, 5 rows affected (0.028 sec)

Records: 5 Duplicates: 0 Warnings: 0

MariaDB [project]> INSERT INTO WorksOn

-> (empNo, projNo, dateWorked, hoursWorked)

-> VALUES

-> (101, 2, '2026-06-01', 20.50),

->

-> (102, 1, '2026-06-02', 18.00),

```

->
-> (103, 4, '2026-06-03', 25.75),
->
-> (104, 5, '2026-06-04', 30.00),
->
-> (105, 3, '2026-06-05', 22.25);
Query OK, 5 rows affected (0.028 sec)
Records: 5 Duplicates: 0 Warnings: 0

```

```

MariaDB [PROJECT]> create view EmpProject(empNo, projNo, totalHours)
-> as select w.empNo, w.projNo, SUM(hoursWorked)
-> from Employee e, Project p, WorksOn w
-> where e.empNo= w.empNo and p.projNo= w.projNo
-> group by w.empNo, w.projNo;

```

```

MariaDB [project]> select *
-> from EmpProject;
+-----+-----+-----+
| empNo | projNo | totalHours |
+-----+-----+-----+
| 101 | 2 | 20.50 |
| 102 | 1 | 18.00 |
| 103 | 4 | 25.75 |
| 104 | 5 | 30.00 |
| 105 | 3 | 22.25 |
+-----+-----+-----+

```

```

MariaDB [project]> select projNo
-> from EmpProject
-> where projNo= 'SCCS';
Empty set, 1 warning (0.012 sec)

```

```

MariaDB [project]> select count(projNo)
-> from EmpProject
-> where empNo= 'E1';
+-----+
| count(projNo) |
+-----+
| 0 |
+-----+

```

```

MariaDB [project]> select empNo, totalHours
-> from EmpProject
-> group by empNo;
+-----+-----+
| empNo | totalHours |
+-----+-----+

```

	101	20.50
	102	18.00
	103	25.75
	104	30.00
	105	22.25

I would first create a table for the view to act upon, then the column would have part_count to track how many contracts qualify for each part. To capture the changes automatically without reading the Part table, I would create 3 triggers to insert, delete and update the table.